MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: MG3SECBOT200 Course Title: BIOFERTILIZERS AND BIOCONTROL AGENTS

Time: 1.5 Hrs Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A:
Short Answers
Answer 10 questions
Each question carries 1 mark

SL No	Question	Taxonomy	CO
1	Define sustainable agriculture.	Remember	1
2	Name one plant-growth promoting rhizobacteria (PGPR).	Remember	2
3	State one benefit of using Azolla in paddy fields.	Remember	2
4	Give one example of a microbial biocontrol agent.	Remember	3
5	What is the active compound present in neem?	Remember	3
6	State one property of a good carrier used in biocontrol formulation	Understand	4
7	What are wettable powders?	Understand	4
8	Name one viral biocontrol agent.	Remember	4
9	What is seed treatment?	Remember	6
10	Name one enzyme involved in composting.	Remember	5
11	What is compost enrichment?	Understand	5
12	What is vermicompost?	Understand	5

Part B:

Short Essay Type Questions
Answer 6 questions
Each question carries 5 marks

13	Summarise the practices of environment-friendly agriculture.	Understand	1
14	Explain the mechanism of egg parasitism by Trichogramma.	Understand	3
15	Examine the ecological benefits of using biocontrol agents in farming.	Analyse	3
16	Explain the need for bioformulations in modern farming.	Understand	4
17	Describe the soil application and soil drenching methods of biofertilizer delivery.	Analyze	4
18	Explain the importance of Rhizobium biofertilizer in sustainable agriculture.	Understand	6
19	Write a note on the environmental and agricultural benefits of household composting.	Understand	5

20	Identify the role of microbial additives in composting.	Apply	5

Part C:

Essay Type Questions

Answer any one question

Question carries 10 marks

21	Classify biofertilizers based on nutrient availability and explain each type with suitable examples.	Understand	2
22	Explain the methods involved in the mass production and formulation of Trichoderma.	Understand	5